# EV Market Adoption Report

### Deep-Dive into Macroeconomic Fuel Spikes, Sourcing Dynamics, and EV Penetration

## 1. Executive Summary & Market Backdrop

Current indicators suggest that the automotive division is experiencing a noticeable transition toward Electrification, although progress remains uneven. Analysis of historical sales databases appears to indicate a general correlation between macroeconomic fuel price shocks and the volume expansion of our Electric Vehicle (EV) portfolio. The fuel price surge of 2022, which saw regional gasoline prices reach approximately $4.65/gallon, may have served as a significant catalyst.

One contributing factor to the success of our dedicated EV brands, particularly "Voltaic", was the growing consumer sensitivity to liquid fuel prices. While conventional gasoline sedans and luxury segments faced transaction resistance due to rising financing rates, EVs maintained a generally positive volume growth of 24% year-over-year. However, it is critical to note that this adoption is highly localized, with some southern and central sub-markets displaying slower EV adoption rates due to charging infrastructure constraints.

## 2. EV Adoption Metrics by Quarter (2019-2023)

| Year | Average Fuel Price ($/Gal) | EV Units Sold | SUV Market Share | Luxury Segment Growth |
|---|---|---|---|---|
| 2019 (Pre-COVID) | $2.45 | 12,450 | 32.1% | +4.5% |
| 2020 (COVID-19) | $2.10 | 14,800 | 35.4% | -8.2% |
| 2021 (COVID-19) | $3.10 | 22,900 | 41.2% | +12.4% |
| 2022 (Inflation) | $4.65 | 41,200 | 48.9% | -18.5% |
| 2023 (Post-Inflation) | $3.85 | 58,400 | 54.2% | -2.1% |

## 3. Regional Infrastructure & Customer Preference

Preliminary analysis shows that charging infrastructure availability remains a major predictor of EV adoption velocity. Our regional sales records indicate that the West region, characterized by robust charging networks, achieved a 35% EV penetration rate in 2023. In contrast, the South region is displaying slower EV adoption, with conventional gasoline and diesel trucks still representing 72% of total dealership sales, which might suggest a geographic divide in purchase behavior.

- West Coast Lead: A dense network of charging stations coupled with local incentives has made EVs a prominent consumer choice.
- Midwest and South Latency: Lack of public DC fast chargers appears to restrict EV sales to urban sub-markets.
- SUV Dominance: Consumers are increasingly demanding larger-chassis vehicles, which may be shifting demand toward Electric SUVs.

## 4. Strategic Recommendations

To exploit these structural shifts and secure a balanced market position, the automotive executive council suggests evaluating the following directives:

- Transition a portion of Sedan assembly lines to Electric SUV platforms by Q3 2024, keeping regional production modular.
- Evaluate shifting ad spend away from luxury internal combustion engine models to highlight EV efficiency and cost savings.
- Collaborate with major regional utility providers in the South and East to support public dealership-based charging projects.

### Macroeconomic Implications of Battery Sourcing

A key risk identified in our operational audit is the supply chain vulnerability of raw battery materials, which has seen an inflationary pressure of 14% over the last fiscal year. This sourcing risk directly affects the manufacturer's suggested retail price (MSRP) of EVs, threatening to widen the affordability gap for middle-income buyers. To mitigate this risk, Stratos and Voltaic must enter into long-term raw materials agreements and invest in domestic recycling partnerships to reclaim up to 25% of lithium and cobalt from end-of-life battery packs.

By pursuing a proactive supply chain strategy alongside localized consumer advertising, the group aims to improve EV gross margins from 12.4% to a target of 18.2% by the end of 2025.